Year 13 Maths - Further transformations of graphs

Similar questions - Modulus equations and inequalities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Solve $|3x-2| = |x+4|$. **[3]**

(b) Hence solve $|3x-2| < |x+4|$. **[2]**

(c) The line $y = x+k$ meets the graph of $y = |3x-2|$ at two distinct points. Find the range of values of k, and the x-coordinates of the points in terms of k. **[5]**

2

(a) Solve $|x-1| \geqslant 2|x+2|$. **[5]**

(b) Solve $|x-4| = 2x-5$, and explain why only one of the algebraic solutions is valid. **[4]**

Question 1

(a) $3x-2=x+4$ gives $x=3$; $3x-2=-(x+4)$ gives $x=-\frac{1}{2}$

(b) The graph of $y=|3x-2|$ is below that of $y=|x+4|$ between the intersections: $-\frac{1}{2}<x<3$

(c) $y=|3x-2|$ is a V with vertex $(\frac{2}{3},0)$ and gradients ± 3. The line has gradient 1, so it meets both arms exactly when the vertex lies below the line: $0<\frac{2}{3}+k$, so $k>-\frac{2}{3}$. Right arm: $3x-2=x+k$ gives $x=\dfrac{k+2}{2}$. Left arm: $2-3x=x+k$ gives $x=\dfrac{2-k}{4}$

Question 2

(a) Squaring both sides (both are non-negative): $(x-1)^2 \geqslant 4(x+2)^2$, so $3x^2+18x+15 \leqslant 0$, i.e. $(x+5)(x+1) \leqslant 0$, giving $-5 \leqslant x \leqslant -1$

(b) $x-4=2x-5$ gives $x=1$; $4-x=2x-5$ gives $x=3$. The right-hand side $2x-5$ must be non-negative, so $x \geqslant 2.5$. $x=1$ gives $|1-4|=3 \neq -3$, so it is rejected. $x=3$: $|3-4|=1=2(3)-5$ ✓